Abstract Algebra II

Lecture notes, Chapters 6–9 (Sections 29–40, 45–56)

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Chapter 6. Extension Fields

References of the form "Theorem 27.25" without a chapter prefix refer to Fraleigh's numbering, used in brackets throughout both volumes.

6.1 Introduction to extension fields (§29)

Motivation. The polynomial x^2+1 has no root in $\mathbb{R}$, so we enlarge $\mathbb{R}$ to $\mathbb{C}$. Kronecker's theorem says this can always be done: every nonconstant polynomial over any field has a root in some larger field, namely $F[x]/\langle p(x)\rangle$. The rest of the chapter studies such extensions $E \supseteq F$, and the central question of Part 9, which polynomials are solvable by radicals, is phrased in their language.

A field E is an **extension field** of F if $F \leq E$ (a subfield).

Theorem 6.1 (29.3, Kronecker)

If F is a field and $f(x) \in F[x]$ is nonconstant, there is an extension field E of F and $\alpha \in E$ with $f(\alpha) = 0$.

Proof. Let $p(x)$ be an irreducible factor of f (Theorem 23.20). $\langle p\rangle$ is maximal (Theorem 27.25), so $E = F[x]/\langle p\rangle$ is a field. $a \mapsto a + \langle p\rangle$ is a one-to-one homomorphism $F \to E$ (its kernel is an ideal of the field F not containing 1), and we identify F with its image. For $\alpha = x + \langle p\rangle$, $p(\alpha) = p(x) + \langle p\rangle = 0$, so $f(\alpha) = 0$. ■

$\alpha \in E$ is **algebraic** over F if $f(\alpha) = 0$ for some nonzero $f \in F[x]$, and **transcendental** otherwise.

Theorem 6.2 (29.12)

Let $\phi_\alpha : F[x] \to E$ be the evaluation homomorphism. α is transcendental over F iff ϕ_α is one-to-one, i.e. an isomorphism of $F[x]$ onto its image (an integral domain).

Proof. $\ker \phi_\alpha = \{f : f(\alpha) = 0\}$ is $\{0\}$ exactly when α is transcendental. ■

Theorem 6.3 (29.13)

If $\alpha \in E$ is algebraic over F, there is an irreducible $p(x) \in F[x]$ with $p(\alpha) = 0$, unique up to a constant factor; it has minimal degree among nonzero polynomials vanishing at α, and $f(\alpha) = 0$ implies $p \mid f$.

Proof. $\ker \phi_\alpha$ is a nonzero ideal of $F[x]$, so equals $\langle p\rangle$ (Theorem 27.24) with p of minimal degree in it. p is irreducible: if $p = gh$ with lower degrees, then $g(\alpha)h(\alpha) = 0$ gives a nonzero polynomial of smaller degree vanishing at α. Every f with $f(\alpha) = 0$ lies in $\langle p\rangle$. If q is another irreducible with $q(\alpha) = 0$, then $p \mid q$, so $q = cp$. ■

The monic such p is $\mathrm{irr}(\alpha, F)$, and its degree is $\deg(\alpha, F)$. $F(\alpha)$ denotes the smallest subfield of E containing F and α (a **simple extension**).

Theorem 6.4 (29.18)

If α is algebraic over F of degree n, every $\beta \in F(\alpha)$ is uniquely $\beta = b_0 + b_1\alpha + \cdots + b_{n-1}\alpha^{n-1}$ with $b_i \in F$.

Proof. $\phi_\alpha[F[x]] \cong F[x]/\langle p\rangle$ is a field (Theorem 27.25), containing F and α, so it equals $F(\alpha)$. Thus $\beta = f(\alpha)$ for some f; dividing, $f = pq + r$ with $\deg r < n$, and $\beta = r(\alpha)$. If two such expressions agree, their difference is a polynomial of degree $< n$ vanishing at α, hence zero. ■

Example 6.1

$\mathbb{R}[x]/\langle x^2+1\rangle \cong \mathbb{R}(i) = \mathbb{C}$. x^2+x+1 is irreducible over $\mathbb{Z}_2$, so $\mathbb{Z}_2(\alpha) = \{0, 1, \alpha, 1+\alpha\}$ with $\alpha^2 = \alpha + 1$ is a field of 4 elements. π and e are transcendental over $\mathbb{Q}$ (Lindemann, Hermite).

6.2 Vector spaces (§30)

Motivation. An extension field E of F is automatically a vector space over F, so its size can be measured by a dimension, the degree $[E : F]$. Linear algebra is reviewed here because degree counting is the main computational tool for the rest of the course.

A **vector space** over F is an abelian group V with a scalar multiplication $F \times V \to V$ satisfying $a(\alpha+\beta) = a\alpha + a\beta$, $(a+b)\alpha = a\alpha + b\alpha$, $a(b\alpha) = (ab)\alpha$, $1\alpha = \alpha$.

Theorem 6.5 (30.5)

In a vector space, $0\alpha = 0$, $a0 = 0$, and $(-a)\alpha = a(-\alpha) = -(a\alpha)$.

Proof. $0\alpha + 0\alpha = (0+0)\alpha = 0\alpha$; cancel. Likewise $a0 = 0$. $a\alpha + (-a)\alpha = 0\alpha = 0$ and $a\alpha + a(-\alpha) = a0 = 0$. ■

Vectors **span** V if every vector is a linear combination of them; they are **linearly independent** if $\sum a_i\alpha_i = 0$ only for all $a_i = 0$; a **basis** is an independent spanning set. V is **finite-dimensional** if a finite set spans it.

Lemma 6.6 (30.16)

If α is a linear combination of the β_i and each β_i is a linear combination of the γ_j, then α is a linear combination of the γ_j.

Proof. $\alpha = \sum_i b_i \sum_j c_{ij}\gamma_j = \sum_j (\sum_i b_i c_{ij})\gamma_j$. ■

Theorem 6.7 (30.17)

Every finite spanning set of a vector space contains a basis.

Proof. Delete from the list $\alpha_1, \ldots, \alpha_n$ every α_i that is a linear combination of the preceding ones. The remaining vectors still span (Lemma 6.6), and they are independent: in a nontrivial relation, the last vector with nonzero coefficient would be a combination of earlier ones. ■

Corollary 6.8 (30.18)

A finite-dimensional vector space has a finite basis.

Theorem 6.9 (30.19)

In a finite-dimensional space, any independent set $\{\alpha_1, \ldots, \alpha_r\}$ can be enlarged to a basis; and if $\{\beta_1, \ldots, \beta_n\}$ is any basis, then $r \le n$.

Proof. Apply the deletion process of Theorem 6.7 to the list $\alpha_1, \ldots, \alpha_r, \beta_1, \ldots, \beta_n$: no α_i is deleted (independence), so the result is a basis containing the α_i. For $r \le n$ (exchange): $\alpha_1 = \sum b_j \beta_j$ with some $b_j \neq 0$, say b_1; then $\alpha_1, \beta_2, \ldots, \beta_n$ is again a basis. Repeating, after k steps $\alpha_1, \ldots, \alpha_k, \beta_{k+1}, \ldots, \beta_n$ is a basis (at step $k+1$, α_{k+1} has a nonzero coefficient on some remaining β, by independence of the α's). If $r > n$, after n steps $\alpha_1, \ldots, \alpha_n$ is a basis and α_{n+1} is a combination of them, a contradiction. ∎

Corollary 6.10 (30.20)

Any two bases of a finite-dimensional vector space have the same number of elements, the **dimension**.

Theorem 6.11 (30.23)

If α is algebraic over F of degree n, then $F(\alpha)$ is an n-dimensional vector space over F with basis $1, \alpha, \ldots, \alpha^{n-1}$, and every $\beta \in F(\alpha)$ is algebraic over F with $\deg(\beta, F) \le n$.

Proof. The basis statement is Theorem 6.4. The $n+1$ elements $1, \beta, \ldots, \beta^n$ are dependent (Theorem 6.9), giving a nonzero polynomial of degree $\le n$ with zero β. ∎

6.3 Algebraic extensions (§31)

Motivation. Degrees multiply in towers, $[K : F] = [K : E][E : F]$. This innocent formula is surprisingly powerful: it shows that algebraic numbers form a field and, in the next section, settles classical construction problems. We also construct the algebraic closure, a field in which every polynomial splits, and prove the fundamental theorem of algebra for $\mathbb{C}$.

E is an **algebraic extension** of F if every element of E is algebraic over F, and a **finite extension** of degree $[E : F]$ if it has finite dimension $[E : F]$ as a vector space over F.

Theorem 6.12 (31.3)

A finite extension is algebraic.

Proof. If $[E : F] = n$, then $1, \alpha, \ldots, \alpha^n$ are dependent for every $\alpha \in E$. ∎

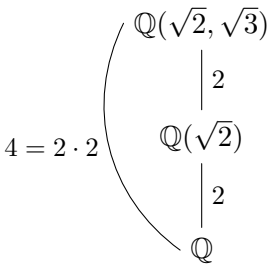


Figure. *Degrees multiply in a tower:* $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})]\,[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 \cdot 2$.

Theorem 6.13 (31.4)

If E is a finite extension of F and K a finite extension of E, then K is a finite extension of F and $[K:F] = [K:E][E:F]$.

Proof. Let $\alpha_1, \ldots, \alpha_n$ be a basis of E over F and $\beta_1, \ldots, \beta_m$ a basis of K over E. Then $\{\alpha_i\beta_j\}$ spans K over F: $\gamma = \sum_j b_j\beta_j = \sum_j \sum_i a_{ij}\alpha_i\beta_j$. It is independent: if $\sum_{i,j} c_{ij}\alpha_i\beta_j = 0$, then $\sum_j(\sum_i c_{ij}\alpha_i)\beta_j = 0$, so each $\sum_i c_{ij}\alpha_i = 0$ and each $c_{ij} = 0$. ■

Corollary 6.14 (31.6)

For a tower $F_1 \leq F_2 \leq \cdots \leq F_r$ of finite extensions, $[F_r : F_1] = [F_r : F_{r-1}] \cdots [F_2 : F_1]$.

Corollary 6.15 (31.7)

If α is algebraic over F and $\beta \in F(\alpha)$, then $\deg(\beta, F)$ divides $\deg(\alpha, F)$.

Proof. $[F(\alpha) : F] = [F(\alpha) : F(\beta)][F(\beta) : F]$ and $[F(\beta) : F] = \deg(\beta, F)$. ■

So $\mathbb{Q}(\sqrt{2})$ contains no $\sqrt[3]{2}$ $(3 \nmid 2)$. $F(\alpha_1, \ldots, \alpha_n)$ is the smallest subfield containing F and the α_i.

Theorem 6.16 (31.11)

An algebraic extension E of F is finitely generated, $E = F(\alpha_1, \ldots, \alpha_n)$, iff it is a finite extension.

Proof. If $E = F(\alpha_1, \ldots, \alpha_n)$, each α_i is algebraic over $F(\alpha_1, \ldots, \alpha_{i-1})$, so each step is finite and $[E:F] < \infty$ (Corollary 6.14). Conversely, a basis over F generates E. ■

Theorem 6.17 (31.12)

For an extension E of F, $\overline{F}_E = \{\alpha \in E : \alpha \text{ algebraic over } F\}$ is a subfield of E (the **algebraic closure of F in E**).

Proof. If $\alpha, \beta \in \overline{F}_E$, then $F(\alpha, \beta)$ is a finite extension (Theorem 6.16), hence algebraic (Theorem 6.12), and contains $\alpha \pm \beta$, $\alpha\beta$ and α/β. ■

Corollary 6.18 (31.13)

The algebraic numbers (complex numbers algebraic over $\mathbb{Q}$) form a field.

F is **algebraically closed** if every nonconstant $f \in F[x]$ has a zero in F; an **algebraic closure** $\overline{F}$ of F is an algebraic extension that is algebraically closed.

Theorem 6.19 (31.15)

F is algebraically closed iff every nonconstant polynomial in $F[x]$ factors into linear factors in $F[x]$.

Proof. If F is algebraically closed, a zero a gives $f = (x-a)g$ (factor theorem); induct on degree. The converse is clear. ■

Corollary 6.20 (31.16)

An algebraically closed field has no proper algebraic extension.

Proof. If α is algebraic over F, $\operatorname{irr}(\alpha, F)$ factors into linear factors over F, so is linear and $\alpha \in F$. ■

Lemma 6.21 (31.21, Zorn's lemma)

If every chain in a nonempty partially ordered set S has an upper bound in S, then S has a maximal element.

Proof. Zorn's lemma is equivalent to the axiom of choice and is taken as an axiom of set theory (see the Topology notes for its equivalence with the well-ordering theorem). ■

Theorem 6.22 (31.17 and 31.22)

Every field F has an algebraic closure $\overline{F}$.

Proof [Supplement]. (Artin.)

Step 1. An extension in which every nonconstant $f \in F[x]$ has a zero. Introduce an indeterminate x_f for each nonconstant $f \in F[x]$, let $R = F[\{x_f\}]$ and I the ideal generated by all $f(x_f)$. $I \neq R$: otherwise $1 = \sum_{i \leq k} g_i f_i(x_{f_i})$; by Kronecker's theorem (Theorem 6.1, repeatedly) there is an extension in which each f_i has a zero α_i, and substituting $x_{f_i} = \alpha_i$ (other variables 0) gives $1 = 0$. By Zorn's lemma, the proper ideals containing I (ordered by inclusion; the union of a chain is proper since it misses 1) have a maximal element M. $E_1 = R/M$ is a field containing F, in which $x_f + M$ is a zero of f; and E_1 is generated by these algebraic elements, so it is algebraic over F (Theorem 6.17).

Step 2. Repeat to get $F \leq E_1 \leq E_2 \leq \cdots$, each algebraic over the previous one, and $L = \bigcup E_n$, a field algebraic over F (algebraic over algebraic is algebraic: if α is a zero of $\sum c_i x^i$ with c_i algebraic over F, then $F(c_0, \ldots, c_m, \alpha)$ is finite over F). Each nonconstant $h \in F[x]$ splits into linear factors over L: it has a zero in E_1, the quotient (with coefficients in E_1) has a zero in E_2, and so on. Now let $g \in L[x]$ be nonconstant, and α a zero of g in some extension K of L (Kronecker). α is algebraic over L, hence over F; let $h = \operatorname{irr}(\alpha, F) = \prod(x - \beta_i)$ with $\beta_i \in L$. As $h(\alpha) = 0$ in K, $\alpha = \beta_i \in L$ for some i. So g has a zero in L, and L is an algebraic closure of F. ■

Theorem 6.23 (31.18, fundamental theorem of algebra)

$\mathbb{C}$ is algebraically closed.

Proof [Supplement]. Let $f \in \mathbb{C}[x]$ be nonconstant of degree n. Since $|f(z)| \to \infty$ as $|z| \to \infty$, $|f|$ attains a minimum at some z_0 (on a large closed disc, by compactness). Suppose $f(z_0) \neq 0$. Expanding, $f(z_0 + w) = f(z_0) + cw^k + w^{k+1}h(w)$ with $c \neq 0$, $k \geq 1$, h a polynomial. Choose $\omega \in \mathbb{C}$ with $\omega^k = -f(z_0)/c$ (k-th roots exist in $\mathbb{C}$). For $0 < t < 1$, $f(z_0 + t\omega) = f(z_0)(1 - t^k) + t^{k+1}\omega^{k+1}h(t\omega)$, so $|f(z_0 + t\omega)| \leq |f(z_0)|(1 - t^k) + Mt^{k+1} < |f(z_0)|$ for small t, where M bounds $|\omega^{k+1}h(t\omega)|$; a contradiction. So $f(z_0) = 0$. ■

A purely algebraic proof via Galois theory and the Sylow theorems is given after Theorem 53.6.

6.4 Geometric constructions (§32)

Motivation. Can one trisect an angle or double a cube with ruler and compass? Constructible numbers lie in towers of quadratic extensions, so their degrees are powers of 2; since $\cos 20^\circ$ and $\sqrt[3]{2}$ have degree 3, both constructions are impossible. Two thousand years of attempts are settled by the tower formula.

A real number α is **constructible** if a segment of length $|\alpha|$ can be constructed with straightedge and compass from a given unit segment.

Theorem 6.24 (32.1)

If α, β are constructible, so are $\alpha \pm \beta$, $\alpha\beta$ and α/β ($\beta \neq 0$).

Proof. Sums and differences: lay off segments on a line. Products and quotients: by similar triangles (on two rays from a point mark 1 and β on one, α on the other; the parallel through β to the line joining 1 and α cuts off $\alpha\beta$, and similarly for α/β). ■

Corollary 6.25 (32.5)

The constructible real numbers form a subfield $\mathcal{F}$ of $\mathbb{R}$ containing $\mathbb{Q}$.

Square roots of positive constructible numbers are constructible (on a segment of length $1+\alpha$ draw a semicircle; the perpendicular at the junction has length $\sqrt{\alpha}$).

Theorem 6.26 (32.6)

$\mathcal{F}$ consists exactly of the real numbers obtainable from $\mathbb{Q}$ by finitely many field operations and square roots of positive numbers.

Proof. Such numbers are constructible by the above. Conversely, all points constructed from points with coordinates in a subfield F of $\mathbb{R}$ lie in F or in $F(\sqrt{a})$, $a \in F$, $a > 0$: intersecting two lines with equations over F gives coordinates in F; intersecting a line and a circle, or two circles (subtracting the equations reduces to a line and a circle), leads to a quadratic equation over F. ■

Corollary 6.27 (32.8)

If γ is constructible and $\gamma \notin \mathbb{Q}$, there are $\gamma_1, \ldots, \gamma_n = \gamma$ with $[\mathbb{Q}(\gamma_1, \ldots, \gamma_i) : \mathbb{Q}(\gamma_1, \ldots, \gamma_{i-1})] = 2$. In particular $[\mathbb{Q}(\gamma) : \mathbb{Q}] = 2^r$.

Proof. By Theorem 6.26 and Corollary 6.15, as $\gamma \in \mathbb{Q}(\gamma_1, \ldots, \gamma_n)$ of degree 2^n. ■

Theorem 6.28 (32.9)

Doubling the cube is impossible.

Proof. It requires constructing $\sqrt[3]{2}$, of degree 3 over $\mathbb{Q}$ ($x^3 - 2$ is irreducible by Eisenstein), not a power of 2. ■

Theorem 6.29 (32.10)

Squaring the circle is impossible.

Proof. For the unit circle it requires $\sqrt{\pi}$, hence π, to be constructible, hence algebraic; but π is transcendental (Lindemann). ■

Theorem 6.30 (32.11)

There is an angle that cannot be trisected; e.g. $60°$.

Proof. Trisecting $60°$ means constructing $\alpha = \cos 20°$. From $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, $8\alpha^3 - 6\alpha - 1 = 0$, i.e. 2α is a zero of $x^3 - 3x - 1$, which has no rational zero (± 1 fail), so is irreducible of degree 3. ■

6.5 Finite fields (§33)

Motivation. Finite fields appear in coding theory, cryptography and number theory. They turn out to be completely classified: for every prime power p^n there is exactly one field of that order up to isomorphism, it is the splitting field of $x^{p^n} - x$, and its multiplicative group is cyclic.

Theorem 6.31 (33.1)

If E is an extension of degree n of a finite field F with q elements, then $|E| = q^n$.

Proof. Each element is uniquely $\sum_{i \le n} b_i \alpha_i$ for a basis α_i. ■

Corollary 6.32 (33.2)

A finite field of characteristic p has p^n elements for some n.

Proof. It is a finite extension of its prime field $\mathbb{Z}_p$. ■

Theorem 6.33 (33.3)

If $E \le \overline{\mathbb{Z}_p}$ has p^n elements, its elements are exactly the zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}_p}$.

Proof. E^* has order $p^n - 1$, so $\alpha^{p^n - 1} = 1$ for $\alpha \neq 0$ and $\alpha^{p^n} = \alpha$ for all $\alpha \in E$. These are p^n zeros, all the zeros of the polynomial of degree p^n. ■

α is a **primitive n-th root of unity** if $\alpha^n = 1$ and $\alpha^m \neq 1$ for $0 < m < n$.

Theorem 6.34 (33.5)

The multiplicative group of a finite field is cyclic.

Proof. Corollary 23.6. ■

Corollary 6.35 (33.6)

A finite extension of a finite field is simple.

Proof. $E = F(\alpha)$ for a generator α of E^*. ■

Lemma 6.36 (33.8)

If F has characteristic p, $x^{p^n} - x$ has p^n distinct zeros in $\overline{F}$.

Proof. It splits in $\overline{F}$; a repeated zero α would be a zero of the derivative $p^n x^{p^n-1} - 1 = -1$, which has no zeros. (Directly: if $f = (x-\alpha)^2 g$, then $f' = (x-\alpha)[2g + (x-\alpha)g']$ vanishes at α.) ■

Lemma 6.37 (33.9)

In characteristic p, $(\alpha + \beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$.

Proof. For $n = 1$, $p \mid \binom{p}{k}$ for $0 < k < p$; induct. ■

Theorem 6.38 (33.10)

For every prime power p^n there is a finite field $\mathrm{GF}(p^n)$ of p^n elements.

Proof. The zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}_p}$ (p^n of them, Lemma 6.36) form a subfield: they are closed under sums (Lemma 6.37), products, negatives and inverses ($(\alpha^{-1})^{p^n} = \alpha^{-1}$). ■

Corollary 6.39 (33.11)

If F is finite, then for every $n \geq 1$ there is an irreducible polynomial of degree n in $F[x]$.

Proof. Let $|F| = p^r$. The zeros of $x^{p^{rn}} - x$ in $\overline{F}$ form a field $E \supseteq F$ (the elements of F satisfy $\alpha^{p^r} = \alpha$) with $[E : F] = n$ (Theorem 6.31). $E = F(\beta)$ by Corollary 6.35, and $\mathrm{irr}(\beta, F)$ has degree n. ■

Theorem 6.40 (33.12)

Any two fields with p^n elements are isomorphic.

Proof. Both are splitting fields of $x^{p^n} - x$ over $\mathbb{Z}_p$ (Theorem 6.33), hence isomorphic (Corollary 49.5 and the uniqueness of splitting fields, Section 9.3). Directly: let $E = \mathbb{Z}_p(\alpha)$ and $f = \mathrm{irr}(\alpha, \mathbb{Z}_p)$; f divides $x^{p^n} - x$, which splits in E', so f has a zero $\alpha' \in E'$, and $\mathbb{Z}_p(\alpha) \cong \mathbb{Z}_p[x]/\langle f \rangle \cong \mathbb{Z}_p(\alpha') \leq E'$; comparing orders, $\mathbb{Z}_p(\alpha') = E'$. ■

Chapter 7. Advanced Group Theory

7.1 Isomorphism theorems (§34)

Motivation. The fundamental homomorphism theorem has several useful consequences that describe how subgroups and factor groups interact. They are the bookkeeping tools for the composition series and Sylow arguments that follow.

Theorem 7.1 (34.2, first isomorphism theorem)

Let $\phi : G \to G'$ be a homomorphism with kernel K and $\gamma_K : G \to G/K$ the canonical map. There is a unique isomorphism $\mu : G/K \to \phi[G]$ with $\phi(x) = \mu(\gamma_K(x))$ for all x.

Proof. Existence is Theorem 14.11.

Uniqueness: $\mu(xK) = \mu(\gamma_K(x))$ must equal $\phi(x)$. ■

Lemma 7.2 (34.3, correspondence)

Let $N \trianglelefteq G$ and $\gamma : G \to G/N$. Then $L \mapsto \gamma[L]$ is a bijection from the normal subgroups of G containing N onto the normal subgroups of G/N.

Proof. $\gamma[L]$ is normal in G/N (Theorem 15.16). It is onto: for $\bar{L} \trianglelefteq G/N$, $L = \gamma^{-1}[\bar{L}]$ is normal, contains N, and $\gamma[L] = \bar{L}$ (γ is onto). One-to-one: if $N \leq L$, then $\gamma^{-1}[\gamma[L]] = LN = L$; so L is recovered from $\gamma[L]$. ■

The same proof shows the correspondence for all subgroups (not only normal ones).

Lemma 7.3 (34.4)

If $N \trianglelefteq G$ and $H \leq G$, then $H \vee N = HN = NH$ (the subgroup generated by $H \cup N$). If moreover $H \trianglelefteq G$, then $HN \trianglelefteq G$.

Proof. $hn = (hnh^{-1})h \in NH$ and similarly $NH \subset HN$. HN is a subgroup: $(h_1n_1)(h_2n_2) = h_1h_2(h_2^{-1}n_1h_2)n_2$ and $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1})$. Any subgroup containing H and N contains HN, so $HN = H \vee N$. If H is also normal, $g(hn)g^{-1} = (ghg^{-1})(gng^{-1}) \in HN$. ■

Theorem 7.4 (34.5, second isomorphism theorem)

If $H \leq G$ and $N \trianglelefteq G$, then $H \cap N \trianglelefteq H$ and $HN/N \cong H/(H \cap N)$.

Proof. $\phi : H \to HN/N$, $\phi(h) = hN$, is a homomorphism; it is onto since $hnN = hN$; its kernel is $H \cap N$. Apply Theorem 7.1. ■

Theorem 7.5 (34.7, third isomorphism theorem)

If $K \leq H$ are normal subgroups of G, then $H/K \trianglelefteq G/K$ and $G/H \cong (G/K)/(H/K)$.

Proof. $\phi : G/K \to G/H$, $\phi(xK) = xH$, is well defined ($K \leq H$), an onto homomorphism, with kernel $\{xK : x \in H\} = H/K$. Apply Theorem 7.1. ■

7.2 Series of groups (§35)

Motivation. Just as integers factor into primes, a finite group can be broken by a chain of normal subgroups into simple factor groups. The Jordan–Hölder theorem says these "prime factors" are unique, and the notion of a solvable group (all factors abelian) is exactly what Galois theory will need for solving equations by radicals.

A **subnormal series** of G is a chain $\{e\} = H_0 < H_1 < \cdots < H_n = G$ with $H_i \trianglelefteq H_{i+1}$; a **normal series** has all $H_i \trianglelefteq G$. A **refinement** inserts further groups. Two series are **isomorphic** if there is a one-to-one correspondence between their factor groups H_{i+1}/H_i with corresponding factors isomorphic. A **composition series** is a subnormal series with all factors simple; a **principal series** is a normal series with no proper normal refinement. G is **solvable** if it has a composition series with abelian (hence cyclic of prime order) factors, equivalently a subnormal series with abelian factors.

Lemma 7.6 (35.10, Zassenhaus)

Let $H, K \leq G$, $H^* \trianglelefteq H$, $K^* \trianglelefteq K$. Then
(1) $H^*(H \cap K^*) \trianglelefteq H^*(H \cap K)$;
(2) $K^*(H^* \cap K) \trianglelefteq K^*(H \cap K)$;
(3) $\dfrac{H^*(H \cap K)}{H^*(H \cap K^*)} \cong \dfrac{K^*(H \cap K)}{K^*(H^* \cap K)} \cong \dfrac{H \cap K}{(H^* \cap K)(H \cap K^*)}$.

Proof. $H \cap K$ normalizes H^* (as $H^* \trianglelefteq H$), so $H^*(H \cap K)$ is a subgroup; similarly the other products. $H^* \cap K \trianglelefteq H \cap K$ (intersect the normal H^* of H with $H \cap K$), likewise $H \cap K^* \trianglelefteq H \cap K$, so $L = (H^* \cap K)(H \cap K^*) \trianglelefteq H \cap K$. Define $\phi : H^*(H \cap K) \to (H \cap K)/L$ by $\phi(hx) = xL$ ($h \in H^*$, $x \in H \cap K$). Well defined: if $hx = h'x'$, then $x'x^{-1} = h'^{-1}h \in H^* \cap (H \cap K) = H^* \cap K \subset L$. Homomorphism: $(h_1x_1)(h_2x_2) = h_1(x_1h_2x_1^{-1})x_1x_2$ with $x_1h_2x_1^{-1} \in H^*$. It is onto, and $\ker\phi = \{hx : x \in L\} = H^*L = H^*(H \cap K^*)$ (since $H^* \cap K \subset H^*$). This proves (1) and the isomorphism of the first and third groups in (3); by symmetry ($H \leftrightarrow K$) we get (2) and the second isomorphism. ∎

Theorem 7.7 (35.11, Schreier)

Two subnormal (normal) series of G have isomorphic refinements.

Proof. Let $\{e\} = H_0 < \cdots < H_n = G$ and $\{e\} = K_0 < \cdots < K_m = G$. Insert $H_{i,j} = H_i(H_{i+1} \cap K_j)$ ($0 \leq j \leq m$) between $H_i = H_{i,0}$ and $H_{i+1} = H_{i,m}$, and $K_{j,i} = K_j(H_i \cap K_{j+1})$ between K_j and K_{j+1}. By Lemma 7.6 (with $H = H_{i+1}$, $H^* = H_i$, $K = K_{j+1}$, $K^* = K_j$), $H_{i,j} \trianglelefteq H_{i,j+1}$, $K_{j,i} \trianglelefteq K_{j,i+1}$, and $H_{i,j+1}/H_{i,j} \cong K_{j,i+1}/K_{j,i}$. Both refined series have nm steps, matched by $(i,j) \leftrightarrow (j,i)$. (Repeated groups give trivial factors, which correspond to each other.) For normal series all groups are normal in G. ∎

Theorem 7.8 (35.15, Jordan–Hölder)

Any two composition (principal) series of G are isomorphic.

Proof. By Theorem 7.7 they have isomorphic refinements. A refinement of a composition series only adds repeated groups (factors are simple, so no proper normal subgroup can be inserted, Theorem 15.18), i.e. trivial factors. Deleting the trivial factors on both sides leaves an isomorphism of the original series. ∎

Theorem 7.9 (35.16)

If G has a composition (principal) series and N is a proper normal subgroup, there is a composition (principal) series containing N.

Proof. The series $\{e\} < N < G$ and the given series have isomorphic refinements. The refinement of the composition series has only simple or trivial factors; so does the isomorphic refinement of $\{e\} < N < G$. Removing repetitions from the latter gives a composition series containing N. ■

Example 7.1

$\{e\} < A_n < S_n$ is a composition series for $n \geq 5$, so S_n is not solvable. The center $Z(G)$ and the ascending central series $Z_0 = \{e\} \leq Z_1 = Z(G) \leq Z_2 \leq \cdots$ $(Z_{i+1}/Z_i = Z(G/Z_i))$ are defined; $\mathbb{Z}_{60}$ has composition factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_5$ in every composition series.

7.3 Sylow theorems (§36)

Motivation. Lagrange's theorem says the order of a subgroup divides $|G|$, but the converse fails (A_4 has no subgroup of order 6). The Sylow theorems give the strongest partial converse: subgroups of every maximal prime-power order exist, are all conjugate, and their number is tightly constrained. The proofs are applications of group actions and orbit counting.

A **p-group** is a group in which every element has order a power of p.

Theorem 7.10 (36.1)

If $|G| = p^n$ and X is a finite G-set, then $|X| \equiv |X_G| \pmod{p}$, where X_G is the set of fixed points.

Proof. X is the disjoint union of orbits; orbits of one element are the fixed points, and every other orbit has size $(G : G_x)$, a divisor of p^n larger than 1 (Theorem 16.16), hence divisible by p. ■

Theorem 7.11 (36.3, Cauchy)

If p is prime and $p \mid |G|$, then G has an element of order p, hence a subgroup of order p.

Proof. Let $X = \{(g_1, \ldots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\}$; $|X| = |G|^{p-1}$ (g_p is determined by the others), divisible by p. $\mathbb{Z}_p$ acts on X by cyclic shifts (if $g_1 \cdots g_p = e$ then $g_2 \cdots g_p g_1 = g_1^{-1} e g_1 = e$). The fixed points are the $(g, \ldots, g)$ with $g^p = e$. By Theorem 7.10 their number is $\equiv 0 \pmod{p}$; it is at least 1 ($g = e$), so there is $g \neq e$ with $g^p = e$. ■

Corollary 7.12 (36.4)

A finite group is a p-group iff its order is a power of p.

Proof. If a prime $q \neq p$ divides $|G|$, Cauchy gives an element of order q. The converse is Lagrange. ■

A **Sylow p-subgroup** is a maximal p-subgroup. The **normalizer** of H is $N[H] = \{g : gHg^{-1} = H\}$, the largest subgroup in which H is normal.

Lemma 7.13 (36.6)

If H is a p-subgroup of a finite group G, then $(N[H]:H) \equiv (G:H) \pmod{p}$.

Proof. H acts on the set $\mathcal{L}$ of left cosets of H by left multiplication. xH is fixed iff $hxH = xH$ for all $h \in H$, iff $x^{-1}hx \in H$ for all h, iff $x \in N[H]$ ($x^{-1}Hx = H$, by finiteness). So the fixed points are the $(N[H]:H)$ cosets of H in $N[H]$; apply Theorem 7.10. ■

Corollary 7.14 (36.7)

If H is a p-subgroup and $p \mid (G:H)$, then $N[H] \neq H$.

Proof. $(N[H]:H) \equiv 0 \pmod{p}$ and is at least 1, so it is at least p. ■

Theorem 7.15 (36.8, first Sylow theorem)

Let $|G| = p^n m$, $n \geq 1$, $p \nmid m$. Then (1) G has a subgroup of order p^i for each $1 \leq i \leq n$; (2) every subgroup H of order p^i, $i < n$, is a normal subgroup of a subgroup of order p^{i+1}.

Proof. Cauchy gives order p. Let $|H| = p^i < p^n$. Then $p \mid (G:H)$, so $p \mid (N[H]:H) = |N[H]/H|$ (Lemma 7.13). By Cauchy, $N[H]/H$ has a subgroup K of order p; its inverse image under $N[H] \to N[H]/H$ has order p^{i+1} and contains H as a normal subgroup. Induction gives (1). ■

So the Sylow p-subgroups of G are exactly the subgroups of order p^n.

Theorem 7.16 (36.10, second Sylow theorem)

Any two Sylow p-subgroups P_1, P_2 of a finite group are conjugate.

Proof. P_1 acts on the left cosets of P_2; their number $(G:P_2)$ is prime to p, so by Theorem 7.10 some xP_2 is fixed: $P_1 x P_2 = xP_2$, so $x^{-1}P_1 x \leq P_2$, with equality by orders. ■

Theorem 7.17 (36.11, third Sylow theorem)

If $p \mid |G|$, the number of Sylow p-subgroups is $\equiv 1 \pmod{p}$ and divides $|G|$.

Proof. Let P be a Sylow subgroup; P acts by conjugation on the set $\mathcal{S}$ of Sylow p-subgroups. If Q is fixed, $P \leq N[Q]$, so P and Q are Sylow subgroups of $N[Q]$, conjugate there (Theorem 7.16); as $Q \trianglelefteq N[Q]$, $P = Q$. So there is exactly one fixed point and $|\mathcal{S}| \equiv 1$. G acts transitively on $\mathcal{S}$ by conjugation (Theorem 7.16), so $|\mathcal{S}| = (G:N[P])$ divides $|G|$. ■

7.4 Applications of the Sylow theory (§37)

Motivation. The Sylow constraints are often enough to determine a group completely or to show that no group of a given order is simple. This section shows the typical arguments, for example that every group of order pq with $p < q$, $q \not\equiv 1 \pmod{p}$, is cyclic.

Theorem 7.18 (37.1)

Every finite p-group is solvable.

Proof. By Theorem 7.15(2) applied repeatedly there is a chain $\{e\} = H_0 < H_1 < \cdots < H_n = G$, $|H_i| = p^i$, $H_i \trianglelefteq H_{i+1}$; the factors have order p, hence are abelian. ■

Theorem 7.19 (37.4)

A finite nontrivial p-group has nontrivial center.

Proof. Let G act on itself by conjugation; the fixed points form the center $Z(G)$. By Theorem 7.10, $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$, and $e \in Z(G)$. ■

Lemma 7.20 (37.5)

If $H, K \trianglelefteq G$, $H \cap K = \{e\}$ and $HK = G$, then $G \cong H \times K$.

Proof. For $h \in H$, $k \in K$, $hkh^{-1}k^{-1} \in H \cap K$ (it is $(hkh^{-1})k^{-1} \in K$ and $h(kh^{-1}k^{-1}) \in H$), so $hk = kh$. Then $(h, k) \mapsto hk$ is a homomorphism $H \times K \to G$, onto, with kernel $\{(h, h^{-1}) : h \in H \cap K\} = \{(e, e)\}$. ■

Theorem 7.21 (37.6)

Every group of order p^2 is abelian.

Proof. $Z(G) \neq \{e\}$. If $Z(G) \neq G$, then $|Z(G)| = p$ and $G/Z(G)$ is cyclic, generated by $aZ(G)$; then every element is $a^i z$ with $z \in Z(G)$, and such elements commute, so G is abelian, $Z(G) = G$, a contradiction. ■

Theorem 7.22 (37.7)

If $p < q$ are primes, a group of order pq has a unique, hence normal, subgroup of order q; it is not simple. If $q \not\equiv 1 \pmod{p}$, then G is cyclic.

Proof. The number of Sylow q-subgroups divides pq, is $\equiv 1 \pmod{q}$, so is 1 (as $p < q$). If $q \not\equiv 1 \pmod{p}$, the number of Sylow p-subgroups, which divides q and is $\equiv 1 \pmod{p}$, is also 1. Both are normal, intersect trivially, and their product has order pq; by Lemma 7.20, $G \cong \mathbb{Z}_p \times \mathbb{Z}_q \cong \mathbb{Z}_{pq}$. ■

Lemma 7.23 (37.8)

For finite subgroups H, K of G, $|HK| = |H||K|/|H \cap K|$.

Proof. HK is the union of the cosets hK, $h \in H$, each of size $|K|$; $hK = h'K$ iff $h^{-1}h' \in H \cap K$, so there are $(H : H \cap K)$ distinct ones. ■

Example 7.2

No group of order 15, 20, 30, 36, 48, or 255 is simple. For order 30: if neither $n_3 = 1$ nor $n_5 = 1$, then $n_3 = 10$, $n_5 = 6$, giving 20 elements of order 3 and 24 of order 5, too many. For order 36: if $n_3 = 4$, the action on the 4 Sylow 3-subgroups gives a homomorphism to S_4 with a nontrivial kernel (as $36 > 24$). For order 48: two Sylow 2-subgroups H, K of order 16 have $|H \cap K| \geq 8$ by Lemma 7.23, and $N[H \cap K]$ contains H and K, so has order 48.

7.5 Free abelian groups (§38)

Motivation. Free abelian groups play the role of vector spaces with a basis among abelian groups. Every finitely generated abelian group is a quotient of $\mathbb{Z}^n$, and choosing good bases for $\mathbb{Z}^n$ and its subgroups gives a full proof of the fundamental theorem stated in §11.

Theorem 7.24 (38.1)

For a subset X of a nonzero abelian group G the following are equivalent:
(1) each nonzero $a \in G$ is uniquely $a = n_1x_1 + \cdots + n_rx_r$ with distinct $x_i \in X$ and $n_i \neq 0$;
(2) X generates G, and $n_1x_1 + \cdots + n_rx_r = 0$ for distinct $x_i \in X$ only if all $n_i = 0$.

Proof. (1)⇒(2): if $\sum n_ix_i = 0$ with some $n_i \neq 0$, then for such an x_j, $x_j = x_j + \sum n_ix_i$ gives two different expressions of x_j (after collecting terms). (2)⇒(1): two expressions of a subtract to a relation, whose coefficients must vanish. ■

Such an X is a **basis**, and G is **free abelian**.

Theorem 7.25 (38.5)

A free abelian group with a basis of r elements is isomorphic to $\mathbb{Z}^r$.

Proof. $\sum n_ix_i \mapsto (n_1, \ldots, n_r)$ is an isomorphism. ■

Theorem 7.26 (38.6)

If $G \neq \{0\}$ is free abelian with a finite basis, every basis is finite and all bases have the same number of elements (the **rank**).

Proof. If X is a basis with r elements, $G/2G \cong \mathbb{Z}_2^r$ has 2^r elements. An infinite basis Y would give infinitely many distinct cosets $y + 2G$ (distinct basis elements are distinct mod $2G$ by uniqueness of coefficients mod 2). ■

Theorem 7.27 (38.8)

If G is generated by $a_1, \ldots, a_n$, then $\phi : \mathbb{Z}^n \to G$, $\phi(h_1, \ldots, h_n) = \sum h_ia_i$, is an onto homomorphism.

Theorem 7.28 (38.9)

If $\{x_1, \ldots, x_r\}$ is a basis of G, $t \in \mathbb{Z}$ and $i \neq j$, then replacing x_j by $x_j + tx_i$ gives another basis.

Proof. The new set generates ($x_j = (x_j + tx_i) - tx_i$), and a relation $\sum n_kx'_k = 0$ becomes $\sum_k n_kx_k + n_jtx_i = 0$, forcing $n_k = 0$ for $k \neq i$ and then $n_i = 0$. ■

Theorem 7.29 (38.11)

If G is free abelian of rank n and $K \neq \{0\}$ a subgroup, then K is free abelian of rank $s \leq n$, and there are a basis $x_1, \ldots, x_n$ of G and positive integers $d_1 \mid d_2 \mid \cdots \mid d_s$ such that $d_1x_1, \ldots, d_sx_s$ is a basis of K.

Proof. This is the stacked-basis argument in the proof of Theorem 11.12 (Volume I, Step 1), carried out with the elementary basis changes of Theorem 7.28. ∎

Theorem 7.30 (38.12)

Every finitely generated abelian group is isomorphic to $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_r} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ with $m_i \geq 2$ and $m_i \mid m_{i+1}$.

Proof. By Theorem 7.27, $G \cong \mathbb{Z}^n/K$. With the bases of Theorem 7.29, $\mathbb{Z}^n/K \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_s} \times \mathbb{Z}^{n-s}$; drop the factors with $d_i = 1$. ∎

Splitting each $\mathbb{Z}_{m_i}$ into prime-power factors gives Theorem 11.12.

7.6 Free groups and presentations (§§39–40)

Motivation. How can one describe an infinite or complicated group finitely? Start with the free group, in which the generators satisfy no relations at all, and impose relations; every group arises this way. Presentations are how groups appear in topology (fundamental groups) and are the natural language for classifying small groups.

For a set A of letters, **words** are finite strings of symbols a, a^{-1} ($a \in A$); **elementary contractions** delete adjacent pairs aa^{-1} or $a^{-1}a$; every word reduces to a unique **reduced word** (uniqueness is checked by van der Waerden's permutation trick, as for free products). The reduced words form a group $F[A]$ under juxtaposition followed by reduction, the **free group** on A. A group G is **free on** $A \subset G$ if $G \cong F[A]$ by an isomorphism fixing A; the cardinality of A is its **rank**.

Theorem 7.31 (39.7)

If G is free on A and on B, then $|A| = |B|$.

Proof. The abelianization G/C (Theorem 15.20) is free abelian with basis the images of A (a homomorphism $G \to \mathbb{Z}^{(A)}$ sending a to the basis vector e_a exists by Theorem 7.34, and kills C), and likewise with basis the images of B. If A is finite, then so is B and $|A| = |B|$ by Theorem 7.26. If both are infinite, $|A| = |F[A]| = |B|$, since the set of finite words over an infinite set has the same cardinality as the set. ∎

Theorem 7.32 (39.9)

Two free groups are isomorphic iff they have the same rank.

Proof. A bijection $A \to A'$ extends to an isomorphism $F[A] \to F[A']$ (rename letters); conversely an isomorphism carries a free generating set to one, and use Theorem 7.31. ∎

Theorem 7.33 (39.10, Nielsen–Schreier)

A nontrivial proper subgroup of a free group is free.

Proof. Realize $F[A]$ as the fundamental group of a wedge of circles X, one for each $a \in A$. For $H \leq F[A]$ there is a covering $E \to X$ with fundamental group isomorphic to H; E is a graph, and the fundamental group of a connected graph is free (generated by the edges outside a maximal tree). Full proofs are in the Topology II notes, Chapter 14. (An algebraic proof uses a Schreier transversal T of H: H is free on the nontrivial elements $ta\,\overline{ta}^{-1}$, $t \in T$, $a \in A$.) ∎

Theorem 7.34 (39.12)

Let G be generated by $A = \{a_i\}$ and let $a_i' \in G'$ be arbitrary. There is at most one homomorphism $\phi : G \to G'$ with $\phi(a_i) = a_i'$; if G is free on A, there is exactly one.

Proof. Every element of G is a product of powers of the a_i (Theorem 7.6), and ϕ is determined on such products. If $G = F[A]$, define $\phi(a_{i_1}^{n_1} \cdots a_{i_k}^{n_k}) = a_{i_1}'^{n_1} \cdots a_{i_k}'^{n_k}$ on reduced words; it is well defined, and since the product of words is computed by juxtaposition and cancellation, which corresponds to valid equalities in G', it is a homomorphism. ∎

Theorem 7.35 (39.13)

Every group is a homomorphic image of a free group.

Proof. Take $A = G$ (or any generating set) and the homomorphism $F[A] \to G$ of Theorem 7.34 extending the identity of A; it is onto. ∎

A **presentation** $(x_j : r_i)$ of G consists of generators x_j and relators r_i (words) such that $G \cong F[\{x_j\}]/N$, N the least normal subgroup containing the r_i. Examples: $\mathbb{Z}_n = (x : x^n)$; $\mathbb{Z} \times \mathbb{Z} = (x, y : xyx^{-1}y^{-1})$; $D_n = (x, y : x^n, y^2, xyxy)$; $Q_8 = (x, y : x^4, x^2y^{-2}, yxy^{-1}x)$. Using Sylow theory and presentations one classifies small groups: for instance there are exactly 5 groups of order 8 ($\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, $\mathbb{Z}_2^3$, D_4, Q_8) and 2 of order 10 ($\mathbb{Z}_{10}$, D_5).

Chapter 8. Factorization

8.1 Unique factorization domains (§45)

Motivation. In $\mathbb{Z}[\sqrt{-5}]$ we have $6 = 2\cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$ with all four factors irreducible, so unique factorization can fail. This chapter asks which integral domains behave like $\mathbb{Z}$. The answer is built in layers: every PID is a UFD, and if D is a UFD then so is $D[x]$, which gives unique factorization in $\mathbb{Z}[x]$ and $F[x_1,\dots,x_n]$.

In an integral domain D: a **divides** b ($a \mid b$) if $b = ac$; a, b are **associates** if $a = bu$ with u a unit; a nonzero nonunit p is **irreducible** if $p = ab$ forces a or b to be a unit. D is a **UFD** if every nonzero nonunit is a product of irreducibles, unique up to order and associates; it is a **PID** if every ideal is principal. Note $\langle a\rangle \subseteq \langle b\rangle$ iff $b \mid a$, and $\langle a\rangle = \langle b\rangle$ iff a, b are associates.

Lemma 8.1 (45.9)

If $N_1 \subseteq N_2 \subseteq \cdots$ are ideals of a commutative ring, then $N = \bigcup N_i$ is an ideal.

Proof. For $a, b \in N$, both lie in some N_i (the chain is increasing), so $a \pm b \in N_i \subseteq N$; and $ra \in N_i$. ■

Lemma 8.2 (45.10, ascending chain condition for a PID)

In a PID every ascending chain of ideals $N_1 \subseteq N_2 \subseteq \cdots$ is eventually constant.

Proof. $\bigcup N_i = \langle c\rangle$ (Lemma 8.1); $c \in N_r$ for some r, and then $N_s \subseteq \langle c\rangle \subseteq N_r \subseteq N_s$ for $s \geq r$. ■

Theorem 8.3 (45.11)

In a PID every nonzero nonunit is a product of irreducibles.

Proof. First, every nonzero nonunit a has an irreducible factor: otherwise $a = a_1b_1$ with a_1, b_1 nonunits, a_1 not irreducible, $a_1 = a_2b_2$, and so on, giving a strictly ascending chain $\langle a\rangle \subsetneq \langle a_1\rangle \subsetneq \cdots$, contradicting Lemma 8.2. Now write $a = p_1c_1$ with p_1 irreducible; if c_1 is not a unit, $c_1 = p_2c_2$, etc. The chain $\langle a\rangle \subsetneq \langle c_1\rangle \subsetneq \langle c_2\rangle \subsetneq \cdots$ must stop, so some c_r is a unit and $a = p_1\cdots p_{r-1}(p_rc_r)$. ■

Lemma 8.4 (45.12)

In a PID, $\langle p\rangle$ is maximal iff p is irreducible.

Proof. As in Theorem 27.25: $\langle p\rangle \subseteq \langle a\rangle$ means $p = ab$; irreducibility says a is a unit ($\langle a\rangle = D$) or b is ($\langle a\rangle = \langle p\rangle$). ■

Lemma 8.5 (45.13)

In a PID, if an irreducible p divides ab, then $p \mid a$ or $p \mid b$.

Proof. $\langle p\rangle$ is maximal, hence prime (Corollary 27.16), and $ab \in \langle p\rangle$. ■

Corollary 8.6 (45.14)

If an irreducible p in a PID divides $a_1 \cdots a_n$, it divides some a_i.

Theorem 8.7 (45.17)

Every PID is a UFD.

Proof. Existence is Theorem 8.3. If $p_1 \cdots p_r = q_1 \cdots q_s$ with irreducibles, p_1 divides some q_j (Corollary 8.6), so $q_j = up_1$ with u a unit; cancel and induct on r (if $r = 1$, the right side is irreducible, so $s = 1$). ■

Corollary 8.8 (45.18, fundamental theorem of arithmetic)

$\mathbb{Z}$ is a UFD.

Proof. Ideals of $\mathbb{Z}$ are the $n\mathbb{Z}$ (Corollary 6.7). ■

Let D be a UFD. A nonconstant $f \in D[x]$ is **primitive** if the only common divisors of its coefficients are units.

Lemma 8.9 (45.23)

Every nonconstant $f \in D[x]$ is $f = cg$ with $c \in D$ and g primitive; c (the **content**) is unique up to a unit.

Proof. Let c be a gcd of the coefficients (it exists in a UFD: take the product of the common irreducible factors with minimal exponents). If $f = cg$ with g primitive, then c is a gcd of the coefficients of f, since in a UFD $\gcd(cb_0, \ldots, cb_n) = c\gcd(b_0, \ldots, b_n)$ up to units and $\gcd(b_i)$ is a unit. So if also $f = c'g'$ with g' primitive, c and c' are both gcds of the same elements, hence associates. ■

Lemma 8.10 (45.25, Gauss's lemma)

In a UFD D, a product of two primitive polynomials is primitive.

Proof. Let p be an irreducible dividing every coefficient of fg. $\langle p \rangle$ is prime in a UFD (if $p \mid ab$, p appears in the factorization of ab), so $D/\langle p \rangle$ is an integral domain and so is $(D/\langle p \rangle)[x]$. The images satisfy $\bar{f}\bar{g} = 0$, so $\bar{f} = 0$ or $\bar{g} = 0$, i.e. p divides all coefficients of f or of g, a contradiction. ■

Corollary 8.11 (45.26)

A finite product of primitive polynomials over a UFD is primitive.

Lemma 8.12 (45.27)

Let F be the field of quotients of a UFD D and $f \in D[x]$ nonconstant. If f is irreducible in $D[x]$, it is irreducible in $F[x]$; and if f is primitive and irreducible in $F[x]$, it is irreducible in $D[x]$.

Proof. Suppose $f = rs$ in $F[x]$ with lower degrees. Clearing denominators, $r = \frac{a}{b}r_1$, $s = \frac{c}{d}s_1$ with $r_1, s_1 \in D[x]$ primitive, so $bdf = ac\, r_1 s_1$. By Lemma 8.9 and Gauss's lemma, the content of the left side is $bd \cdot \mathrm{cont}(f)$ and that of the right side is ac; so $f = u\ \mathrm{cont}(f)\, r_1 s_1$ with u a unit, a factorization

in $D[x]$ into lower degrees. For the second statement, a factorization in $D[x]$ of a primitive f into nonunits must have both factors of positive degree (a constant factor would divide all coefficients), hence is a factorization in $F[x]$. ■

Corollary 8.13 (45.28)

For D a UFD with field of quotients F, a nonconstant $f \in D[x]$ factors into polynomials of degrees r, s in $F[x]$ iff it does in $D[x]$.

Theorem 8.14 (45.29)

If D is a UFD, so is $D[x]$.

Proof. Let $f \in D[x]$ be a nonzero nonunit, $f = cg$ with g primitive. c factors into irreducibles of D, which remain irreducible in $D[x]$. g factors in the UFD $F[x]$ into irreducibles; by the argument of Lemma 8.12 the factors may be taken primitive in $D[x]$, with product g up to a unit; they are irreducible in $D[x]$.

Uniqueness: in two factorizations, the primitive factors of positive degree give factorizations of the primitive part in $F[x]$, unique up to constants in F^*; two primitive polynomials that are associates in $F[x]$ differ by a unit of D (compare contents). The constant factors give factorizations of the content, unique in D. ■

Corollary 8.15 (45.30)

$F[x_1, \dots, x_n]$ is a UFD for a field F.

Proof. Induction, using Theorem 8.14 and $F[x_1, \dots, x_n] = F[x_1, \dots, x_{n-1}][x_n]$. ■

A UFD need not be a PID: $\langle x, y\rangle \subset F[x, y]$ and $\langle 2, x\rangle \subset \mathbb{Z}[x]$ are not principal.

8.2 Euclidean domains (§46)

Motivation. The reason $\mathbb{Z}$ and $F[x]$ have unique factorization is the division algorithm. A Euclidean domain is any integral domain with a division algorithm measured by a size function; every such domain is a PID, hence a UFD, and gcd's can be computed by the Euclidean algorithm.

A **Euclidean norm** on D is $\nu : D - \{0\} \to \{0, 1, 2, \dots\}$ with (1) for $b \neq 0$ there are q, r with $a = bq + r$, $r = 0$ or $\nu(r) < \nu(b)$;

(2) $\nu(a) \leq \nu(ab)$ for $a, b \neq 0$. Examples: $|n|$ on $\mathbb{Z}$; deg on $F[x]$.

Theorem 8.16 (46.4)

Every Euclidean domain is a PID.

Proof. For a nonzero ideal N choose $b \in N$ with $\nu(b)$ minimal. For $a \in N$, $a = bq + r$ with $r \in N$; minimality forces $r = 0$. So $N = \langle b\rangle$. ■

Corollary 8.17 (46.5)

A Euclidean domain is a UFD.

Theorem 8.18 (46.6)

$\nu(1)$ is minimal among the $\nu(a)$, $a \neq 0$, and u is a unit iff $\nu(u) = \nu(1)$.

Proof. $\nu(1) \leq \nu(1a) = \nu(a)$. If u is a unit, $\nu(u) \leq \nu(uu^{-1}) = \nu(1)$. Conversely, if $\nu(u) = \nu(1)$, write $1 = uq + r$; $r \neq 0$ would give $\nu(r) < \nu(u) = \nu(1)$, impossible; so $uq = 1$. ■

Theorem 8.19 (46.9, Euclidean algorithm)

For nonzero a, b in a Euclidean domain, define $a = bq_1 + r_1$, $b = r_1q_2 + r_2$, $r_1 = r_2q_3 + r_3, \ldots$ with $\nu(r_{i+1}) < \nu(r_i)$. Some $r_s = 0$, and r_{s-1} (or b if $r_1 = 0$) is a gcd of a and b.

Proof. The $\nu(r_i)$ strictly decrease, so the process stops. Each step preserves the set of common divisors ($\langle a, b\rangle = \langle b, r_1\rangle = \ldots$, Theorem 28.6), and $\langle r_{s-1}, 0\rangle = \langle r_{s-1}\rangle$; so $\langle a, b\rangle = \langle r_{s-1}\rangle$, and r_{s-1} divides a, b and is a linear combination of them. ■

8.3 Gaussian integers and multiplicative norms (§47)

Motivation. The Gaussian integers $\mathbb{Z}[i]$ are a Euclidean domain, and factoring in $\mathbb{Z}[i]$ answers a classical question about $\mathbb{Z}$: an odd prime p is a sum of two squares exactly when $p \equiv 1 \pmod 4$. Multiplicative norms are the tool that transfers information between the two rings.

$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, with norm $N(a + bi) = a^2 + b^2$.

Lemma 8.20 (47.2)

For $\alpha, \beta \in \mathbb{Z}[i]$:
(1) $N(\alpha) \geq 0$;
(2) $N(\alpha) = 0$ iff $\alpha = 0$;
(3) $N(\alpha\beta) = N(\alpha)N(\beta)$.

Proof. $N(\alpha) = \alpha\bar{\alpha} = |\alpha|^2$, and $|\alpha\beta| = |\alpha||\beta|$. ■

Lemma 8.21 (47.3)

$\mathbb{Z}[i]$ is an integral domain.

Proof. It is a subring of the field $\mathbb{C}$ containing 1. ■

Theorem 8.22 (47.4)

$\nu(\alpha) = N(\alpha)$ is a Euclidean norm on $\mathbb{Z}[i]$.

Proof. (2): $N(\alpha) \leq N(\alpha)N(\beta)$ as $N(\beta) \geq 1$. (1): given α, $\beta \neq 0$, write $\alpha/\beta = r + si \in \mathbb{Q}(i)$ and choose integers q_1, q_2 with $|r - q_1|, |s - q_2| \leq \frac{1}{2}$. With $q = q_1 + q_2 i$, $\rho = \alpha - \beta q = \beta(\alpha/\beta - q)$ has $N(\rho) = N(\beta)|\alpha/\beta - q|^2 \leq N(\beta)(\frac{1}{4} + \frac{1}{4}) < N(\beta)$. ■

A **multiplicative norm** on an integral domain D is $N : D \to \mathbb{Z}$ with $N(\alpha) = 0$ iff $\alpha = 0$ and $N(\alpha\beta) = N(\alpha)N(\beta)$.

Theorem 8.23 (47.7)

If N is a multiplicative norm on D, then $N(1) = 1$ and $|N(u)| = 1$ for units u. If moreover every α with $|N(\alpha)| = 1$ is a unit, then every π with $|N(\pi)| = p$ prime is irreducible.

Proof. $N(1) = N(1)^2 \neq 0$ gives $N(1) = 1$; $N(u)N(u^{-1}) = 1$ in $\mathbb{Z}$. If $\pi = \alpha\beta$, then $|N(\alpha)||N(\beta)| = p$, so one factor has norm ± 1 and is a unit. ■

Example 8.1

In $\mathbb{Z}[\sqrt{-5}]$ with $N(a+b\sqrt{-5}) = a^2+5b^2$, $6 = 2\cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$, and $2, 3, 1\pm\sqrt{-5}$ are irreducible (no elements of norm 2 or 3); so $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

Theorem 8.24 (47.10, Fermat)

An odd prime p is a^2+b^2 with $a, b \in \mathbb{Z}$ iff $p \equiv 1 \pmod 4$.

Proof. Squares are $\equiv 0, 1 \pmod 4$, so $a^2+b^2 \not\equiv 3$. Conversely let $p \equiv 1 \pmod 4$. $\mathbb{Z}_p^*$ is cyclic of order $p-1$, divisible by 4, so it has an element n of order 4, and $n^2 = -1$: $p \mid n^2+1 = (n+i)(n-i)$. p divides neither $n+i$ nor $n-i$ in $\mathbb{Z}[i]$ (as $\frac{n}{p} \pm \frac{1}{p} i \notin \mathbb{Z}[i]$), so p is not irreducible in the UFD $\mathbb{Z}[i]$ (Lemma 45.13 and Theorem 8.22): $p = (a+bi)(c+di)$ with nonunits. Taking norms, $p^2 = (a^2+b^2)(c^2+d^2)$ with both factors > 1, so $p = a^2+b^2$. ■

Chapter 9. Automorphisms and Galois Theory

Throughout, $\overline{F}$ is a fixed algebraic closure of F and all algebraic extensions of F are taken inside it.

9.1 Automorphisms of fields (§48)

Motivation. Galois's idea was to study a polynomial through the symmetries of its roots. Concretely, these are the automorphisms of a field extension that fix the base field; they permute the roots of each polynomial over the base field. This section collects the basic examples, conjugation isomorphisms and the Frobenius map.

α, β algebraic over F are **conjugate over** F if $\operatorname{irr}(\alpha, F) = \operatorname{irr}(\beta, F)$.

Theorem 9.1 (48.3, conjugation isomorphisms)

If α, β are algebraic over F with $\deg(\alpha, F) = n$, the map $\psi_{\alpha,\beta}(c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1}) = c_0 + c_1\beta + \cdots + c_{n-1}\beta^{n-1}$ is an isomorphism $F(\alpha) \to F(\beta)$ iff α and β are conjugate over F.

Proof. If $\psi_{\alpha,\beta}$ is an isomorphism, it fixes F, so $\operatorname{irr}(\alpha, F)(\beta) = \psi(\operatorname{irr}(\alpha, F)(\alpha)) = 0$, hence $\operatorname{irr}(\beta, F) = \operatorname{irr}(\alpha, F)$. Conversely, if $p = \operatorname{irr}(\alpha, F) = \operatorname{irr}(\beta, F)$, the evaluation maps give isomorphisms $F[x]/\langle p\rangle \to F(\alpha)$ and $F[x]/\langle p\rangle \to F(\beta)$ (Theorem 29.18); $\psi_{\alpha,\beta}$ is the composite of the inverse of the first with the second. ■

Corollary 9.2 (48.5)

An isomorphism of $F(\alpha)$ onto a subfield of $\overline{F}$ fixing F maps α to a conjugate of α; conversely, for each conjugate β there is exactly one such isomorphism with $\alpha \mapsto \beta$, namely $\psi_{\alpha,\beta}$.

Proof. $\psi(\alpha)$ is a zero of $\operatorname{irr}(\alpha, F)$; and an isomorphism fixing F is determined by the image of α. ■

Corollary 9.3 (48.6)

If $f \in \mathbb{R}[x]$ and $f(a + bi) = 0$, then $f(a - bi) = 0$.

Proof. Complex conjugation is $\psi_{i,-i} : \mathbb{R}(i) \to \mathbb{R}(i)$, fixing the coefficients of f. ■

An isomorphism of E onto itself is an **automorphism**. For automorphisms σ_i, the **fixed field** is $E_{\{\sigma_i\}} = \{a : \sigma_i(a) = a\ \forall i\}$.

Theorem 9.4 (48.11)

$E_{\{\sigma_i\}}$ is a subfield of E.

Proof. Each σ_i preserves sums, products, negatives and inverses. ■

Theorem 9.5 (48.14)

The automorphisms of a field form a group under composition.

Theorem 9.6 (48.15)

For $F \leq E$, the automorphisms of E fixing F form a subgroup $G(E/F)$, whose fixed field contains F.

Proof. Composites and inverses of automorphisms fixing F fix F. ■

Theorem 9.7 (48.19, Frobenius)

For a finite field F of characteristic p, $\sigma_p(a) = a^p$ is an automorphism of F with fixed field $\mathbb{Z}_p$.

Proof. σ_p is a ring homomorphism (Lemma 33.9), one-to-one ($a^p = 0$ only for $a = 0$), hence onto the finite set F. The fixed elements are the zeros of $x^p - x$, i.e. the p elements of $\mathbb{Z}_p$ (Fermat). ■

9.2 The isomorphism extension theorem (§49)

Motivation. To count automorphisms we need to know when an isomorphism between small fields extends to larger ones. The isomorphism extension theorem guarantees such extensions exist, and the number of ways of extending (the index $\{E : F\}$) becomes the key invariant of the next sections.

Theorem 9.8 (49.3, isomorphism extension theorem)

Let E be algebraic over F, $\sigma : F \to F'$ an isomorphism, and $\overline{F'}$ an algebraic closure of F'. Then σ extends to an isomorphism τ of E onto a subfield of $\overline{F'}$.

Proof. Consider pairs (L, λ) with $F \leq L \leq E$ and $\lambda : L \to \overline{F'}$ an isomorphism onto its image extending σ, ordered by extension. The union of a chain is again such a pair, so Zorn's lemma gives a maximal (K, λ). If $K \neq E$, take $\alpha \in E - K$ and $p = \operatorname{irr}(\alpha, K)$; the polynomial $\lambda(p)$ is irreducible over $\lambda[K]$ and has a zero $\beta \in \overline{F'}$. As in Theorem 9.1, $K(\alpha) \cong K[x]/\langle p \rangle \cong \lambda[K][x]/\langle \lambda(p) \rangle \cong \lambda[K](\beta)$ gives an extension of λ to $K(\alpha)$, contradicting maximality. So $K = E$. ■

Corollary 9.9 (49.4)

If $E \leq \overline{F}$ is algebraic over F and $\alpha, \beta \in E$ are conjugate over F, then $\psi_{\alpha,\beta}$ extends to an isomorphism of E onto a subfield of $\overline{F}$.

Corollary 9.10 (49.5)

Any two algebraic closures of F are isomorphic by an isomorphism fixing F.

Proof. Extend id_F to $\tau : \overline{F} \to \overline{F}'$. $\tau[\overline{F}]$ is algebraically closed, and $\overline{F}'$ is algebraic over it, so $\overline{F}' = \tau[\overline{F}]$ (Corollary 31.16). ■

The same argument shows: every isomorphism of $\overline{F}$ into itself fixing F is an automorphism; and every isomorphism $\sigma : F \to F'$ extends to an isomorphism $\overline{F} \to \overline{F'}$.

Theorem 9.11 (49.7)

Let E be a finite extension of F, $\sigma : F \to F'$ an isomorphism and $\overline{F'}$ an algebraic closure of F'. The number of extensions of σ to an isomorphism of E onto a subfield of $\overline{F'}$ is finite, independent of σ, F' and $\overline{F'}$, and at most $[E : F]$.

Proof. First let $E = F(\alpha)$, $p = \operatorname{irr}(\alpha, F)$. Extensions of σ correspond bijectively to the zeros of $\sigma(p)$ in $\overline{F'}$ (an extension sends α to such a zero; conversely each zero β gives one, as in the proof of Theorem 9.8). Extending σ to an isomorphism $\overline{F} \to \overline{F'}$ (remark after Corollary 9.10) carries the zeros of p bijectively to those of $\sigma(p)$, so their number m equals the number of distinct zeros of p in $\overline{F}$; $m \le \deg p = [F(\alpha) : F]$. In general, write $E = F(\alpha_1, \dots, \alpha_n)$; each extension of σ to $F(\alpha_1, \dots, \alpha_{k-1})$ has the same number m_k of extensions to $F(\alpha_1, \dots, \alpha_k)$, by the simple case; so the total is $m_1 \cdots m_n \le \prod[F(\alpha_1, \dots, \alpha_k) : F(\alpha_1, \dots, \alpha_{k-1})] = [E : F]$. ■

The number is the **index** $\{E : F\}$.

Corollary 9.12 (49.10)

If $F \le E \le K$ with K finite over F, then $\{K : F\} = \{K : E\}\{E : F\}$.

Proof. Each of the $\{E : F\}$ extensions of id_F to E has exactly $\{K : E\}$ extensions to K (independence in Theorem 9.11). ■

9.3 Splitting fields (§50)

Motivation. For Galois theory to work, an extension must contain all the conjugates of each of its elements, not just some of them; $\mathbb{Q}(\sqrt[3]{2})$ fails this. Splitting fields are exactly the extensions with this property, and for them every embedding into an algebraic closure is an automorphism.

E with $F \le E \le \overline{F}$ is a **splitting field over** F if it is the smallest subfield of $\overline{F}$ containing F and all zeros in $\overline{F}$ of each polynomial of some set $\{f_i\} \subseteq F[x]$.

Theorem 9.13 (50.3)

E ($F \le E \le \overline{F}$) is a splitting field over F iff every automorphism of $\overline{F}$ fixing F maps E onto itself.

Proof. If E is generated by the zeros of the f_i, an automorphism τ fixing F permutes the zeros of each f_i, so $\tau[E] \subseteq E$, and likewise $\tau^{-1}[E] \subseteq E$. Conversely, suppose the condition holds; we show E is the splitting field of $\{\operatorname{irr}(\alpha, F) : \alpha \in E\}$, i.e. every conjugate β of every $\alpha \in E$ lies in E. $\psi_{\alpha,\beta}$ extends to an isomorphism of $\overline{F}$ into itself fixing F (Theorem 9.8), which is an automorphism (remark after Corollary 9.10); it maps E onto E, so $\beta = \tau(\alpha) \in E$. ■

Corollary 9.14 (50.6)

If $E \le \overline{F}$ is a splitting field over F, every irreducible polynomial in $F[x]$ with a zero in E splits in E.

Proof. Its zeros are the conjugates of the given zero, which lie in E by the proof of Theorem 9.13. ■

Corollary 9.15 (50.7)

If $E \leq \overline{F}$ is a splitting field over F, every isomorphism of E onto a subfield of $\overline{F}$ fixing F is an automorphism of E. If E is a finite splitting field, $\{E : F\} = |G(E/F)|$.

Proof. Such an isomorphism extends to an automorphism of $\overline{F}$ (as above), which maps E onto E. ■

Example 9.1

$\mathbb{Q}(\sqrt[3]{2}, \omega)$ ($\omega = e^{2\pi i/3}$) is the splitting field of $x^3 - 2$, of degree 6; $\mathbb{Q}(\sqrt[3]{2})$ is not a splitting field over $\mathbb{Q}$. The splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$ is $\mathrm{GF}(p^n)$.

9.4 Separable extensions (§51)

Motivation. For the number of automorphisms to equal the degree, irreducible polynomials must not have repeated roots. Separability is this condition; it holds automatically in characteristic 0 and over finite fields, and it yields the primitive element theorem $E = F(\alpha)$.

α is a zero of f of **multiplicity** ν if $(x - \alpha)^\nu$ divides f in $\overline{F}[x]$ but $(x - \alpha)^{\nu+1}$ does not.

Theorem 9.16 (51.2)

All zeros in $\overline{F}$ of an irreducible $f \in F[x]$ have the same multiplicity.

Proof. For zeros α, β, $\psi_{\alpha,\beta}$ extends to an automorphism τ of $\overline{F}$ fixing F. If $f = (x - \alpha)^\nu g$ with $g(\alpha) \neq 0$, applying τ to coefficients, $f = (x - \beta)^\nu \tau(g)$ with $\tau(g)(\beta) = \tau(g(\alpha)) \neq 0$; by unique factorization in $\overline{F}[x]$ the multiplicity of β is ν. ■

Corollary 9.17 (51.3)

An irreducible $f \in F[x]$ factors in $\overline{F}[x]$ as $a \prod_i (x - \alpha_i)^\nu$ with the α_i its distinct zeros.

Theorem 9.18 (51.6)

If E is finite over F, then $\{E : F\}$ divides $[E : F]$.

Proof. For $E = F(\alpha)$: $[F(\alpha) : F] = \deg \mathrm{irr}(\alpha, F) = \nu \cdot m$ with $m = \{F(\alpha) : F\}$ the number of distinct zeros (Corollary 9.17, Theorem 9.11). In general use a chain of simple extensions, Theorem 31.4 and Corollary 9.12. ■

E is a **separable extension** of F if $\{E : F\} = [E : F]$; α is **separable** over F if $F(\alpha)$ is, i.e. if $\mathrm{irr}(\alpha, F)$ has distinct zeros.

Theorem 9.19 (51.9)

For finite $F \leq E \leq K$: K is separable over F iff K is separable over E and E is separable over F.

Proof. $\{K : F\} = \{K : E\}\{E : F\}$ and $[K : F] = [K : E][E : F]$ with $\{K : E\} \leq [K : E]$ and $\{E : F\} \leq [E : F]$; equality of the products holds iff both equalities hold. ■

Corollary 9.20 (51.10)

A finite extension E of F is separable iff every $\alpha \in E$ is separable over F.

Proof. If E is separable, apply Theorem 9.19 to $F \leq F(\alpha) \leq E$. Conversely let $E = F(\alpha_1, \ldots, \alpha_n)$; $\operatorname{irr}(\alpha_i, F(\alpha_1, \ldots, \alpha_{i-1}))$ divides $\operatorname{irr}(\alpha_i, F)$, so has distinct zeros, and each step is separable; apply Theorem 9.19. ■

Lemma 9.21 (51.11)

If $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \overline{F}[x]$, $f^m \in F[x]$ and $m \cdot 1 \neq 0$ in F, then $f \in F[x]$.

Proof. Suppose $a_{n-1}, \ldots, a_{n-k+1} \in F$. The coefficient of x^{nm-k} in f^m is ma_{n-k} plus a polynomial expression in $a_{n-1}, \ldots, a_{n-k+1}$ with integer coefficients; it lies in F, so $ma_{n-k} \in F$ and $a_{n-k} \in F$. Induct on k. ■

A field is **perfect** if every finite extension is separable.

Theorem 9.22 (51.13)

Every field of characteristic 0 is perfect.

Proof. Let f be monic irreducible over F with distinct zeros α_i of multiplicity ν, and $g = \prod(x - \alpha_i)$. Then $g^\nu = f \in F[x]$ and $\nu \cdot 1 \neq 0$, so $g \in F[x]$ (Lemma 9.21); $g \mid f$ and f irreducible force $\nu = 1$. So every algebraic element is separable; apply Corollary 9.20. ■

Theorem 9.23 (51.14)

Every finite field is perfect.

Proof. If F is finite and α is algebraic over F, then $F(\alpha)$ is finite, of p^n elements, so α is a zero of $x^{p^n} - x$, which has distinct zeros (Lemma 33.8); $\operatorname{irr}(\alpha, F)$ divides it, so has distinct zeros. ■

Theorem 9.24 (51.15, primitive element theorem)

A finite separable extension E of F is simple: $E = F(\gamma)$.

Proof. If F is finite, use Corollary 33.6. Let F be infinite; by induction on the number of generators it suffices to treat $E = F(\alpha, \beta)$. Let $\sigma_1, \ldots, \sigma_n$ be the $n = \{E : F\} = [E : F]$ distinct isomorphisms of E into $\overline{F}$ fixing F. For $i \neq j$, $(\sigma_i\alpha, \sigma_i\beta) \neq (\sigma_j\alpha, \sigma_j\beta)$, so $\sigma_i\alpha + c\sigma_i\beta = \sigma_j\alpha + c\sigma_j\beta$ for at most one $c \in F$. Choose c outside these finitely many values and put $\gamma = \alpha + c\beta$. The σ_i restricted to $F(\gamma)$ are n distinct isomorphisms (they differ on γ), so $[F(\gamma) : F] \geq \{F(\gamma) : F\} \geq n = [E : F]$, and $F(\gamma) = E$. ■

Corollary 9.25 (51.16)

A finite extension of a field of characteristic 0 is simple.

9.5 Totally inseparable extensions (§52)

Motivation. In characteristic p, separability can fail, for instance for $x^p - t$ over $\mathbb{Z}_p(t)$. This section shows that the bad behaviour can be isolated: every algebraic extension is a separable extension

followed by a totally inseparable one.

A finite extension E of F is **totally inseparable** if $\{E : F\} = 1 < [E : F]$; $\alpha \notin F$ is totally inseparable if $F(\alpha)$ is.

Theorem 9.26 (52.3)

For finite $F < E < K$: K is totally inseparable over F iff K is over E and E is over F.

Proof. $\{K : F\} = \{K : E\}\{E : F\}$ is 1 iff both factors are 1. ■

Corollary 9.27 (52.4)

A finite extension $E > F$ is totally inseparable iff every $\alpha \in E - F$ is totally inseparable over F.

Proof. If E is totally inseparable, then $\{E : F\} = \{E : F(\alpha)\}\{F(\alpha) : F\} = 1$ forces $\{F(\alpha) : F\} = 1$, while $[F(\alpha) : F] > 1$. Conversely if $E = F(\alpha_1, \ldots, \alpha_n)$ with each $\mathrm{irr}(\alpha_i, F)$ having one distinct zero, then so does $\mathrm{irr}(\alpha_i, F(\alpha_1, \ldots, \alpha_{i-1}))$, which divides it; so every step has index 1, and $\{E : F\} = 1$. ■

Theorem 9.28 (52.5)

Let F have characteristic $p \neq 0$ and E be finite over F. Then $\alpha \in E - F$ is totally inseparable over F iff $\alpha^{p^t} \in F$ for some $t \geq 1$. Moreover the separable elements of E form a subfield K (the **separable closure** of F in E), and E is totally inseparable over K unless $E = K$.

Proof. If $\alpha^{p^t} = a \in F$, then $x^{p^t} - a = (x - \alpha)^{p^t}$ has the single zero α, so $\mathrm{irr}(\alpha, F)$ does too. Conversely, if $\mathrm{irr}(\alpha, F) = (x - \alpha)^m$, write $m = p^t k$ with $p \nmid k$; then $(x - \alpha)^m = (x^{p^t} - \alpha^{p^t})^k$, and Lemma 9.21 gives $x^{p^t} - \alpha^{p^t} \in F[x]$; $t \geq 1$ since otherwise $k = m$ and $x - \alpha \in F[x]$.

Separable elements form a subfield by Corollary 9.20 applied to $F(\alpha, \beta)$. For $\alpha \in E$, $f = \mathrm{irr}(\alpha, F)$ can be written $g(x^{p^t})$ with g irreducible and $g' \neq 0$ (if $f' = 0$ then $f \in F[x^p]$; iterate), and then g has distinct zeros ($\gcd(g, g') = 1$), so α^{p^t} is separable, i.e. lies in K. By the first part (over K), E is totally inseparable over K. ■

9.6 Galois theory (§53)

Motivation. This is the culmination of the course. For a finite normal (splitting and separable) extension, intermediate fields correspond one-to-one to subgroups of the Galois group, with inclusions reversed and normal subgroups matching normal extensions. Questions about fields become questions about finite groups, which are much easier to compute with.

A finite extension K of F is a **finite normal extension** if it is a separable splitting field over F; then $G(K/F)$ is its **Galois group**, and $|G(K/F)| = \{K : F\} = [K : F]$ (Corollary 9.15).

Theorem 9.29 (53.2)

If K is finite normal over F and $F \leq E \leq K$, then K is finite normal over E, and $G(K/E)$ is the subgroup of $G(K/F)$ of automorphisms fixing E. Two automorphisms $\sigma, \tau \in G(K/F)$ induce the same isomorphism of E into $\overline{F}$ iff they lie in the same left coset of $G(K/E)$.

Proof. K is separable over E (Theorem 9.19), and the splitting field over F of a set of polynomials is also their splitting field over E. $\sigma|_E = \tau|_E$ iff $\sigma^{-1}\tau$ fixes E. ■

Theorem 9.30 (53.6, main theorem of Galois theory)

Let K be finite normal over F, $G = G(K/F)$, and for $F \le E \le K$ let $\lambda(E) = G(K/E)$. Then:
(1) λ is a bijection from the intermediate fields onto the subgroups of G;
(2) $\lambda(E) = G(K/E)$ and $E = K_{G(K/E)}$;
(3) for $H \le G$, $\lambda(K_H) = H$;
(4) $[K : E] = |\lambda(E)|$ and $[E : F] = (G : \lambda(E))$;
(5) E is normal over F iff $\lambda(E) \trianglelefteq G$, and then $G(E/F) \cong G/\lambda(E)$;
(6) λ reverses inclusions.

Proof. (2) Clearly $E \subseteq K_{G(K/E)}$. If $\alpha \in K - E$, $\mathrm{irr}(\alpha, E)$ has degree > 1 and distinct zeros, all in K (Corollary 9.14); for a zero $\beta \neq \alpha$, $\psi_{\alpha,\beta}$ extends to an automorphism of K fixing E (Theorem 9.8 and Corollary 9.15), which moves α. So $E = K_{G(K/E)}$, and λ is one-to-one.

(3) Let $E = K_H$; then $H \le \lambda(E)$. K is separable over E, so $K = E(\alpha)$ (Theorem 9.24). The polynomial $\prod_{\sigma\in H}(x - \sigma\alpha)$ has coefficients fixed by H, i.e. in E, and has α as a zero, so $[K : E] = \deg(\alpha, E) \le |H|$. But $|\lambda(E)| = |G(K/E)| = [K : E]$. Hence $H = \lambda(E)$; this proves surjectivity in (1).

(4) $|\lambda(E)| = [K : E]$, and $[E : F] = [K : F]/[K : E] = |G|/|\lambda(E)|$.

(5) For $\sigma \in G$, $\lambda(\sigma[E]) = \sigma\lambda(E)\sigma^{-1}$. E is a splitting field over F iff every isomorphism of E into $\overline{F}$ fixing F maps E onto E (Theorem 9.13); these isomorphisms are restrictions of elements of G (extend to K, which is a splitting field); so E is normal iff $\sigma[E] = E$ for all $\sigma \in G$, iff $\sigma\lambda(E)\sigma^{-1} = \lambda(E)$ for all σ (by (1)). Then restriction $G \to G(E/F)$ is a homomorphism, onto (every automorphism of E over F extends to K), with kernel $\lambda(E)$.

(6) $E_1 \subseteq E_2$ implies $G(K/E_2) \le G(K/E_1)$, and conversely by (2). ■

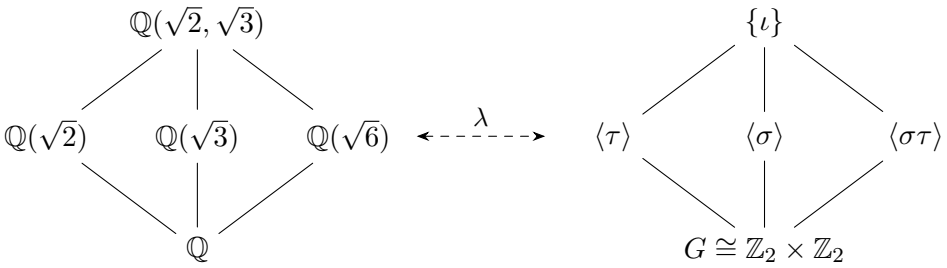


Figure. *The Galois correspondence for $K = \mathbb{Q}(\sqrt{2},\sqrt{3})$, with $\sigma : \sqrt{2} \mapsto -\sqrt{2}$ and $\tau : \sqrt{3} \mapsto -\sqrt{3}$. Each intermediate field is the fixed field of the subgroup in the same position; the lattices are inverted (larger fields correspond to smaller groups).*

Theorem 9.31 (53.7)

If K is an extension of degree n of a finite field F of p^r elements, then $G(K/F)$ is cyclic of order n, generated by $\sigma_{p^r}(\alpha) = \alpha^{p^r}$.

Proof. K is the splitting field of $x^{p^{rn}} - x$ over F, separable (Theorem 9.23), so normal with $|G(K/F)| = n$. σ_{p^r} fixes F (whose elements satisfy $a^{p^r} = a$). If $\sigma_{p^r}^m = \mathrm{id}$ with $0 < m < n$, every $\alpha \in K$ would be a zero of $x^{p^{rm}} - x$, which has fewer than p^{rn} zeros. So σ_{p^r} has order n. ■

Example 9.2

For $K=\mathbb{Q}(\sqrt{2},\sqrt{3})$, $G(K/\mathbb{Q})\cong\mathbb{Z}_2\times\mathbb{Z}_2$, and the three subgroups of order 2 correspond to $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$. For $K=\mathbb{Q}(\sqrt[3]{2},\omega)$, $G\cong S_3$; the normal subgroup A_3 corresponds to the normal extension $\mathbb{Q}(\omega)$.

Algebraic proof of the fundamental theorem of algebra. Let K be a finite normal extension of $\mathbb{R}$ containing i, with $G=G(K/\mathbb{R})$ of order $2^k m$, m odd. The fixed field of a Sylow 2-subgroup has odd degree m over $\mathbb{R}$; by the intermediate value theorem every real polynomial of odd degree has a real zero, so $m=1$. Then $G(K/\mathbb{C})$ is a 2-group; if nontrivial it has a subgroup of index 2, giving a quadratic extension of $\mathbb{C}$, impossible since every complex number has a square root. So $K=\mathbb{C}$.

9.7 Illustrations of Galois theory (§54)

Motivation. The correspondence is best understood through examples: computing the Galois group of a splitting field and drawing the subgroup and subfield lattices side by side. Symmetric functions also appear here as the fixed field of the full symmetric group.

For indeterminates $y_1,\dots,y_n$ the **elementary symmetric functions** s_k are defined by $\prod(x-y_i)=x^n-s_1x^{n-1}+s_2x^{n-2}-\cdots+(-1)^n s_n$.

Theorem 9.32 (54.2)

Every symmetric function of $y_1,\dots,y_n$ over F is a rational function of $s_1,\dots,s_n$: $F(y_1,\dots,y_n)^{S_n}=F(s_1,\dots,s_n)$.

Proof. Let $K=F(y_1,\dots,y_n)$ and $E=F(s_1,\dots,s_n)\le K$. K is the splitting field over E of $\prod(x-y_i)\in E[x]$, which has distinct zeros, so K is finite normal over E with $[K:E]\le n!$ (adjoin the zeros one at a time). S_n acts on K by permuting the y_i, by automorphisms fixing E; so $|G(K/E)|\ge n!$, hence $=n!$, and $G(K/E)=S_n$. By Theorem 9.30(2), $E=K_{S_n}$, the field of symmetric functions. ∎

Example 9.3

$y_1^2+\cdots+y_n^2=s_1^2-2s_2$. The Galois group of x^4-2 over $\mathbb{Q}$ is the dihedral group D_4 of order 8, and its lattice of 10 subgroups matches the lattice of intermediate fields of $\mathbb{Q}(\sqrt[4]{2},i)$.

9.8 Cyclotomic extensions (§55)

Motivation. Adjoining roots of unity produces abelian Galois groups, and these extensions are the first step of every radical tower. As an application we obtain Gauss's characterization of the regular polygons constructible with ruler and compass.

The splitting field of x^n-1 over F is the **n-th cyclotomic extension**. Over $\mathbb{Q}$ it is $\mathbb{Q}(\zeta)$ with $\zeta=e^{2\pi i/n}$; the primitive n-th roots of unity are the ζ^k with $\gcd(k,n)=1$, zeros of the **cyclotomic polynomial** $\Phi_n(x)=\prod_{\gcd(k,n)=1}(x-\zeta^k)$, which lies in $\mathbb{Z}[x]$ (from $x^n-1=\prod_{d\mid n}\Phi_d$ by induction).

Theorem 9.33 (55.4)

The Galois group of the n-th cyclotomic extension of $\mathbb{Q}$ has $\phi(n)$ elements and is isomorphic to the group G_n of units of $\mathbb{Z}_n$.

Proof.

Step 1. [Supplement] (Φ_n is irreducible over $\mathbb{Q}$.) Let $f = \operatorname{irr}(\zeta, \mathbb{Q})$; by Gauss's lemma $f \in \mathbb{Z}[x]$ and $x^n - 1 = fg$ with $g \in \mathbb{Z}[x]$ monic. Let p be a prime not dividing n; we show $f(\zeta^p) = 0$. Otherwise $g(\zeta^p) = 0$, so ζ is a zero of $g(x^p)$ and $f(x) \mid g(x^p)$ in $\mathbb{Z}[x]$. Reduce modulo p: $\bar{g}(x^p) = \bar{g}(x)^p$ in $\mathbb{Z}_p[x]$, so $\bar{f} \mid \bar{g}^p$ and $\bar{f}, \bar{g}$ have a common zero in $\overline{\mathbb{Z}_p}$. Then $x^n - \bar{1} = \bar{f}\bar{g}$ has a repeated zero; but its derivative nx^{n-1} is nonzero at every zero ($p \nmid n$), a contradiction. Applying this repeatedly to the prime factors of k, every ζ^k with $\gcd(k, n) = 1$ is a zero of f; hence $f = \Phi_n$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \phi(n)$.

Step 2. An automorphism σ is determined by $\sigma(\zeta)$, a primitive root ζ^k with $k \in G_n$; $\sigma \mapsto k$ is a homomorphism ($\sigma\tau(\zeta) = \sigma(\zeta^l) = \zeta^{kl}$), one-to-one, and onto since both groups have $\phi(n)$ elements. ■

Corollary 9.34 (55.6)

For a prime p, the Galois group of the p-th cyclotomic extension of $\mathbb{Q}$ is cyclic of order $p - 1$.

Proof. $G_p = \mathbb{Z}_p^*$ is cyclic (Corollary 23.6). ■

Theorem 9.35 (55.8, Gauss)

The regular n-gon is constructible with straightedge and compass iff all odd primes dividing n are Fermat primes ($2^{2^k} + 1$) whose squares do not divide n.

Proof. The n-gon is constructible iff $\cos(2\pi/n)$ is constructible. Since ζ is a zero of $x^2 - 2\cos(2\pi/n)x + 1$, $[\mathbb{Q}(\zeta) : \mathbb{Q}(\cos 2\pi/n)] = 2$ for $n \geq 3$. If $\cos(2\pi/n)$ is constructible, its degree is a power of 2 (Corollary 32.8), so $\phi(n)$ is a power of 2. Conversely, if $\phi(n) = 2^m$, the Galois group G_n is an abelian 2-group; the subgroup fixing $\mathbb{Q}(\cos 2\pi/n)$ has order 2, and the quotient, an abelian group of order 2^{m-1}, has a chain of subgroups each of index 2 (Theorem 36.8). By the Galois correspondence this gives a tower $\mathbb{Q} = L_0 < L_1 < \cdots < L_{m-1} = \mathbb{Q}(\cos 2\pi/n)$ of real fields with $[L_{j+1} : L_j] = 2$; each $L_{j+1} = L_j(\sqrt{d_j})$ with $0 < d_j \in L_j$, so $\cos(2\pi/n)$ is constructible (Theorem 32.6). Finally $\phi(n) = \prod p^{a-1}(p-1)$ is a power of 2 iff every odd $p \mid n$ has $a = 1$ and $p - 1$ a power of 2; and $2^s + 1$ prime forces s to be a power of 2. ■

So the regular 17-gon is constructible, and the regular 7-gon and 9-gon are not.

9.9 Insolvability of the quintic (§56)

Motivation. We can now answer the question that motivated Galois. A polynomial is solvable by radicals exactly when its Galois group is solvable; since S_5 is not solvable (A_5 is simple), there are quintic polynomials over $\mathbb{Q}$ whose roots cannot be written using $+, -, \times, \div$ and radicals.

An extension K of F is an **extension by radicals** if $K = F(\alpha_1, \ldots, \alpha_r)$ with $\alpha_i^{n_i} \in F(\alpha_1, \ldots, \alpha_{i-1})$; $f \in F[x]$ is **solvable by radicals** if its splitting field lies in such a K. Fields have characteristic 0 in this section.

Lemma 9.36 (56.3)

If K is the splitting field of $x^n - a$ over F, then $G(K/F)$ is solvable.

Proof. If $a = 0$ there is nothing to show. The n zeros of $x^n - a$ are distinct (the derivative nx^{n-1} does not vanish at them); if α is one, the quotients α'/α are n distinct n-th roots of unity in K, forming a cyclic group (Corollary 23.6) with a generator ζ, a primitive n-th root of unity. So $K = F(\zeta, \alpha)$. $F(\zeta)$ is normal over F with $G(F(\zeta)/F)$ isomorphic to a subgroup of G_n, abelian. An element of $G(K/F(\zeta))$ is determined by $\alpha \mapsto \zeta^k \alpha$, and $\sigma \mapsto k$ is a one-to-one homomorphism into $\mathbb{Z}_n$; so $G(K/F(\zeta))$ is abelian. The series $\{e\} \leq G(K/F(\zeta)) \trianglelefteq G(K/F)$ has abelian factors (Theorem 9.30(5)). ■

Theorem 9.37 (56.4)

If $F \leq E \leq K$ with E normal over F and K an extension of F by radicals, then $G(E/F)$ is solvable.

Proof. Let $K = F(\alpha_1, \ldots, \alpha_r)$ as above, $n = \prod n_i$, and L the splitting field over F of $(x^n - 1)\prod_i \operatorname{irr}(\alpha_i, F)$; $L \supseteq K \supseteq E$ and L is normal over F. L is generated over $F(\zeta)$ (ζ a primitive n-th root of unity) by the images $\sigma(\alpha_i)$, $\sigma \in G(L/F)$, and $\sigma(\alpha_i)^{n_i} \in F(\sigma(\alpha_1), \ldots, \sigma(\alpha_{i-1}))$. Ordering these generators accordingly gives a tower $F \leq F(\zeta) = L_0 \leq L_1 \leq \cdots \leq L_s = L$ with $L_j = L_{j-1}(\beta_j)$, $\beta_j^{m_j} \in L_{j-1}$, $m_j \mid n$. Since L_{j-1} contains the m_j-th roots of unity, L_j is the splitting field of $x^{m_j} - \beta_j^{m_j}$ over L_{j-1}, normal with abelian Galois group (proof of Lemma 9.36). Then $G(L/L_s) \leq G(L/L_{s-1}) \leq \cdots \leq G(L/L_0) \leq G(L/F)$ is a subnormal series (Theorem 9.30(5)) whose factors $G(L_j/L_{j-1})$ and $G(F(\zeta)/F)$ are abelian. So $G(L/F)$ is solvable, and so is its factor group $G(E/F) \cong G(L/F)/G(L/E)$ (factor groups of solvable groups are solvable, by Theorem 34.5 applied to a series). ■

Theorem 9.38 (56.6)

If $y_1, \ldots, y_5$ are independent transcendentals over $\mathbb{Q}$, then $f(x) = \prod_{i=1}^{5}(x - y_i)$ is not solvable by radicals over $F = \mathbb{Q}(s_1, \ldots, s_5)$.

Proof. By the proof of Theorem 9.32, the splitting field $\mathbb{Q}(y_1, \ldots, y_5)$ has Galois group S_5 over F. S_5 is not solvable: its composition series $\{e\} < A_5 < S_5$ has the nonabelian simple factor A_5 (Theorem 15.15, Jordan–Hölder). By Theorem 9.37, f is not solvable by radicals. ■

Consequently there is no general formula by radicals for the zeros of a polynomial of degree 5; concretely, $x^5 - 4x + 2$ (irreducible by Eisenstein, with exactly three real zeros, so its Galois group contains a 5-cycle and a transposition and is S_5) is not solvable by radicals over $\mathbb{Q}$.